

Chapter 4: Definite Integrals

National High School of Mathematics

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1. Definite Integrals

1.1 The Riemann Integral

Now, we give the definition of the Riemann integral of a function.

Definition 9.1

Let $[a, b]$ be a closed interval.

1. The set $P = \{x_0, x_1, x_2, \dots, x_n\}$ of finite points $a = x_0, x_1, x_2, \dots, x_n = b$ with

$$a = x_0 < x_1 < x_2 < \dots < x_n = b$$

is called a **partition** of $[a, b]$;

2. For every $i = 1, 2, \dots, n$, a closed interval $I_i = [x_{i-1}, x_i]$ is called the **subinterval** of $[a, b]$;
3. For every $i = 1, 2, \dots, n$, the length of I_i is defined by $\Delta x_i = x_i - x_{i-1}$,
4. $\wp[a, b]$ is denoted by the set of all partitions of $[a, b]$,
5. The width of the largest sub-interval in a partition is called the **norm** of the partition.

1.1.1 Step functions

Definition 9.2

A function $f : [a, b] \rightarrow \mathbb{R}$ is called a **step function** if it is piecewise constant, i.e. if there are numbers $a = x_0 < x_1 < x_2 < \dots < x_{N-1} < x_N = b$ such that f is constant on each half open interval $[x_{i-1}, x_i)$ with $1 \leq i \leq N$. For a step function we define the integral to be $\int_a^b f(x)dx = \sum_{i=1}^N f(x_{i-1})(x_i - x_{i-1})$. The collection of numbers $x_0, x_1, x_2, \dots, x_N$ are called a partition for the step function f .

In this definition we have to take into account that a step function f might be defined using different partitions, and show that the integral does not depend on which partition is used to compute it.

Lemma 9.1

If f and g are step functions on an interval $[a, b]$ with $f(x) \leq g(x)$ for all $x \in [a, b]$, then $\int_a^b f(x)dx \leq \int_a^b g(x)dx$.

Definition 9.3

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function and $P = \{a = x_0, x_1, x_2, \dots, x_n = b\}$ be a partition of $[a, b]$. For every subinterval $I_i = [x_{i-1}, x_i]$. We define **Riemann sum** $S(f, P) = \sum_{i=1}^n f(x_i^*)\Delta x_i$, where $\Delta x_i = x_i - x_{i-1}$ and $x_i^* \in [x_{i-1}, x_i]$.

Remark 9.1

One might produce different Riemann sums depending on which x_i^* 's are chosen. In the end this will not matter, if the function is Riemann integrable, when the difference or width of the summands Δx_i approaches zero.

1.1.2 Types of Riemann sums

Specific choices of x_i^* give different types of Riemann sums:

1. If $x_i^* = x_{i-1}$ for all i , the method is the **left rule** and gives a left Riemann sum.
2. If $x_i^* = x_i$ for all i , the method is the **right rule** and gives Riemann sum.
3. If $x_i^* = (x_i + x_{i-1})/2$ for all i , the method is the **midpoint rule** and gives a middle Riemann sum.
4. If $f(x_i^*) = \sup f([x_{i-1}, x_i])$ (that is, the supremum of f over $[x_{i-1}, x_i]$), the method is the upper rule and gives an **upper Riemann sum** or **upper Darboux sum**.
5. If $f(x_i^*) = \inf f([x_{i-1}, x_i])$ (that is, the infimum of f over $[x_{i-1}, x_i]$), the method is the lower rule and gives a **lower Riemann sum** or **lower Darboux sum**.

Definition 9.4

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function and $P = \{a = x_0, x_1, x_2, \dots, x_n = b\}$ be a partition of $[a, b]$. For every subinterval $I_i = [x_{i-1}, x_i]$, put

$$M_i = \sup\{f(x) : x \in [x_{i-1}, x_i]\}, \quad m_i = \inf\{f(x) : x \in [x_{i-1}, x_i]\}$$

Then, for every $P \in \wp[a, b]$, define the **upper Riemann sum** $U(f, P)$ and the **lower**

Riemann sum $L(f, P)$ as follows, respectively:

$$U(f, P) = \sum_{i=1}^n M_i \Delta x_i, \quad L(f, P) = \sum_{i=1}^n m_i \Delta x_i$$

Example 9.1

Let $f : [a, b] \rightarrow \mathbb{R}$ be a function defined by $f(x) = c$, where $c \in \mathbb{R}$ is a constant. Prove that, for every $P = \{a = x_0, x_1, x_2, \dots, x_n = b\} \in \wp[a, b]$

$$U(f, P) = L(f, P) = c(b - a)$$

Solution. Let $\Delta x_i = x_i - x_{i-1}$ for every $i = 1, 2, \dots, n$. Since $M_i = m_i = c$, we have the following:

$$U(f, P) = \sum_{i=1}^n M_i \Delta x_i = \sum_{i=1}^n c \Delta x_i = c \sum_{i=1}^n \Delta x_i = c(b - a)$$

and

$$L(f, P) = \sum_{i=1}^n m_i \Delta x_i = \sum_{i=1}^n c \Delta x_i = c \sum_{i=1}^n \Delta x_i = c(b - a).$$

Thus, we have

$$U(f, P) = L(f, P) = c(b - a)$$

Example 9.2

Let $f : [0, 1] \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \notin \mathbb{Q}^c \end{cases}$$

Then, calculate $U(f, P)$ and $L(f, P)$.

Solution. Let $P = \{0 = x_0, x_1, x_2, \dots, x_n = 1\} \in \wp[0, 1]$. Each subinterval $I_i = [x_{i-1}, x_i]$ for every $i = 1, 2, \dots, n$ has infinitely many rational numbers and irrational numbers, and so, for every $i = 1, 2, \dots, n$,

$$M_i = \sup\{f(x) : x \in [x_{i-1}, x_i]\} = 1$$

and

$$m_i = \inf\{f(x) : x \in [x_{i-1}, x_i]\} = 0.$$

Hence, we have $L(f, P) = 0$ and $U(f, P) = 1$.

Lemma 9.2

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function and $P = \{a = x_0, x_1, x_2, \dots, x_n = b\}$ be a partition of $[a, b]$. Then, we have the following: (1) $L(f, P) \leq U(f, P)$; (2) $m(b - a) \leq L(f, P) \leq U(f, P) \leq M(b - a)$, where $m, M \in \mathbb{R}$ are numbers such that

$$m \leq f(x) \leq M \text{ for every } x \in [a, b]$$

Definition 9.5

Let $[a, b]$ be a closed interval.

1. For every $P, Q \in \wp[a, b]$, Q is called the **refinement** of P if

$$P \subset Q$$

2. The partition $R = P \cup Q$ is called the **common refinement** of P and Q .

Lemma 9.3

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function. For every $P, Q \in \wp[a, b]$, if Q is a refinement of P , then we have the following: (1) $L(f, P) \leq L(f, Q)$
(2) $U(f, Q) \leq U(f, P)$.

• Proof

- (1) First, let

$$P = \{a = x_0, x_1, x_2, \dots, x_n = b\}$$

and

$$Q = \{a = x_0, x_1, x_2, \dots, x_{j-1}, x^*, x_j, \dots, x_n = b\}$$

that is, $Q = P \cup \{x^*\}$. Then, Q is a refinement of P , that is, $P \subset Q$. Next, put

$$m'_j = \inf\{f(x) : x \in [x_{j-1}, x^*]\}, \quad m''_j = \inf\{f(x) : x \in [x^*, x_j]\}$$

and

$$m_j = \min\{m'_j, m''_j\}$$

Then, we have

$$L(f, P) = \sum_{i=1}^n m_i \Delta x_i = \sum_{i=1, i \neq j}^n m_i \Delta x_i + m_j \Delta x_j$$

and

$$L(f, Q) = \sum_{i=1}^{j-1} m_i \Delta x_i + m'_j(x^* - x_{j-1}) + m''_j(x_j - x^*) + \sum_{i=j+1}^n m_i \Delta x_i.$$

Thus, since $m_j \leq m'_j, m''_j$, we have

$$m_j \Delta x_j = m_j(x^* - x_{j-1}) + m_j(x_j - x^*) \leq m'_j(x^* - x_{j-1}) + m''_j(x_j - x^*)$$

and so

$$\sum_{i=1, i \neq j}^n m_i \Delta x_i = \sum_{i=1, i \neq j}^n m_i(x^* - x_{i-1}) + \sum_{i=1, i \neq j}^n m_i(x_i - x^*) \leq \sum_{i=1, i \neq j}^n m'_i(x^* - x_{i-1}) + \sum_{i=1, i \neq j}^n m''_i(x_i - x^*),$$

which implies that $L(f, P) \leq L(f, Q)$. Next, let Q be the refinement adjoining a finite points to P , and then, repeating the proceeding process, we can prove $L(f, P) \leq L(f, Q)$. (2) By the same argument as in (1), we can prove $U(f, Q) \leq U(f, P)$. This completes the proof. $\square$

Theorem 9.4

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function. For every $P, Q \in \wp[a, b]$, (Q isn't a refinement of P), we have $L(f, P) \leq U(f, Q)$.

• Proof

Let $T = P \cup Q$. Then, T is a refinement of P and Q . Thus, by Lemma (9.3), we have

$$L(f, P) \leq L(f, T), \quad U(f, T) \leq U(f, Q).$$

Therefore, since $L(f, P) \leq U(f, P)$, we have

$$L(f, P) \leq U(f, Q).$$

This completes the proof. $\square$

Corollary 9.1

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function. For every $P, Q \in \wp[a, b]$, if $L(f, P) \leq U(f, Q)$, then

$$\sup\{L(f, P) : P \in \wp[a, b]\}, \quad \inf\{U(f, P) : P \in \wp[a, b]\}$$

exist.

• **Proof**

Let $Q \in \wp[a, b]$ be fixed. Then, for every $P \in \wp[a, b]$, we have

$$L(f, P) \leq U(f, Q), \quad L(f, Q) \leq U(f, P),$$

which means that $U(f, Q)$ and $L(f, Q)$ are upper bound and lower bound of the following sets:

$$\{L(f, P) : P \in \wp[a, b]\}, \quad \{U(f, P) : P \in \wp[a, b]\},$$

respectively. Therefore, by the Completeness Axiom of $\mathbb{R}$, it follows that

$$\sup\{L(f, P) : P \in \wp[a, b]\}, \quad \inf\{U(f, P) : P \in \wp[a, b]\}$$

exist. This completes the proof. □

Corollary 9.2

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be bounded functions and $c \in \mathbb{R}$. For every $P \in \wp[a, b]$, we have the following:

1. $U(f + g, P) \leq U(f, P) + U(g, P)$
2. $L(f + g, P) \geq L(f, P) + L(g, P);$
3. If $c \geq 0$, then we have

$$U(cf, P) = cU(f, P), \quad L(cf, P) = cL(f, P)$$

4. If $c < 0$, then we have

$$U(cf, P) = cL(f, P), \quad L(cf, P) = cU(f, P).$$

• **Proof**

Let $P = \{a = x_0, x_1, x_2, \dots, x_n = b\} \in \wp[a, b]$. (1) We have

$$\begin{aligned} U(f + g, P) &= \sum_{i=1}^n \sup\{f(x) + g(x) : x \in [x_{i-1}, x_i]\} \Delta x_i \\ &\leq \sum_{i=1}^n [\sup\{f(x) : x \in [x_{i-1}, x_i]\} + \sup\{g(x) : x \in [x_{i-1}, x_i]\}] \Delta x_i \\ &= U(f, P) + U(g, P) \end{aligned}$$

(2) As in the proof of (1), we have $L(f + g, P) \geq L(f, P) + L(g, P)$. (3) Let $c \geq 0$.

Then, we have

$$U(cf, P) = \sum_{i=1}^n \sup\{cf(x) : x \in [x_{i-1}, x_i]\} \Delta x_i = c \sum_{i=1}^n \sup\{f(x) : x \in [x_{i-1}, x_i]\} \Delta x_i = cU(f, P)$$

(4) Let $c < 0$. Then, we have

$$\begin{aligned} U(cf, P) &= \sum_{i=1}^n \sup\{cf(x) : x \in [x_{i-1}, x_i]\} \Delta x_i \\ &= c \sum_{i=1}^n \inf\{f(x) : x \in [x_{i-1}, x_i]\} \Delta x_i \\ &= cL(f, P). \end{aligned}$$

This completes the proof. □

Definition 9.6

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function.

1. The **upper Riemann integral** of f on $[a, b]$ is the number

$$\int_a^{\overline{b}} f(x) dx = \inf\{U(f, P) : P \in \wp[a, b]\}$$

2. The **lower Riemann integral** of f on $[a, b]$ is the number

$$\int_a^{\underline{b}} f(x) dx = \sup\{L(f, P) : P \in \wp[a, b]\}$$

3. f is said to be **Riemann integrable** on $[a, b]$ if

$$\int_a^{\overline{b}} f(x) dx = \int_a^{\underline{b}} f(x) dx$$

4. In this case, the **Riemann integral** of f is defined to be the number

$$\int_a^b f(x) dx = \int_a^{\overline{b}} f(x) dx = \int_a^{\underline{b}} f(x) dx$$

which is denoted by $\int_a^b f(x) dx$.

5. In addition, we define

$$\int_b^a f(x) dx = - \int_a^b f(x) dx, \quad \int_a^a f(x) dx = 0$$

6. f is said to be not Riemann integrable on $[a, b]$ if

$$\int_a^{\overline{b}} f(x) dx \neq \int_a^{\underline{b}} f(x) dx$$

Remark 9.2

Note that $\mathfrak{R}[a, b]$ denotes the set of Riemann integrable functions on $[a, b]$. From the definition of the Riemann integral, it follows that, for every $P, Q \in \wp[a, b]$ with $P \subset Q$

$$L(f, P) \leq L(f, Q) \leq \int_a^b f(x)dx \leq U(f, Q) \leq U(f, P)$$

Example 9.3

Let $f : [a, b] \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \notin \mathbb{Q}^c \end{cases}$$

Then, we have $L(f, P) = 0$ and $U(f, P) = 1$ and so

$$\overline{\int_a^b} f(x)dx = \inf\{U(f, P) : P \in \wp[a, b]\} = 1$$

and

$$\underline{\int_a^b} f(x)dx = \sup\{L(f, P) : P \in \wp[a, b]\} = 0.$$

Therefore, since $\overline{\int_a^b} f(x)dx = 1 \neq 0 = \underline{\int_a^b} f(x)dx$, f is not Riemann integrable on $[0, 1]$.

Theorem 9.5 (The Riemann Integral Test)

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function. Then, the following are equivalent: (1)

$f \in \mathfrak{R}[a, b]$

(2) For every $\varepsilon > 0$, there exists $P \in \wp[a, b]$ such that

$$U(f, P) - L(f, P) < \varepsilon$$

• Proof

(1) $\implies$ (2) Suppose that $f \in \mathfrak{R}[a, b]$. Since

$$\int_a^b f(x)dx = \inf\{U(f, P) : P \in \wp[a, b]\}$$

for every $\varepsilon > 0$, $\int_a^b f(x)dx + \frac{\varepsilon}{2}$ is not a lower bound of $\{U(f, P) : P \in \wp[a, b]\}$, and so there exists $P_1 \in \wp[a, b]$ such that

$$U(f, P_1) < \int_a^b f(x)dx + \frac{\varepsilon}{2}$$

Also, since

$$\int_a^b f(x)dx = \sup\{L(f, P) : P \in \wp[a, b]\}$$

$\int_a^b f(x)dx - \frac{\varepsilon}{2}$ is not an upper bound of $\{L(f, P) : P \in \wp[a, b]\}$, and so there exists $P_2 \in \wp[a, b]$ such that

$$\int_a^b f(x)dx - \frac{\varepsilon}{2} < L(f, P_2)$$

Letting $P = P_1 \cup P_2$, we have

$$L(f, P_2) \leq L(f, P) \leq U(f, P) \leq U(f, P_1)$$

Thus, we have

$$\begin{aligned} U(f, P) - L(f, P) &\leq U(f, P_1) - L(f, P_2) \\ &< \left(\int_a^b f(x)dx + \frac{\varepsilon}{2} \right) - \left(\int_a^b f(x)dx - \frac{\varepsilon}{2} \right) = \varepsilon. \end{aligned}$$

(2) $\implies$ (1) Suppose that, for every $\varepsilon > 0$, there exists $P \in \wp[a, b]$ such that

$$U(f, P) - L(f, P) < \varepsilon$$

Since $\overline{\int_a^b} f(x)dx \leq U(f, P)$ and $L(f, P) \leq \underline{\int_a^b} f(x)dx$, we have

$$0 \leq \overline{\int_a^b} f(x)dx - \underline{\int_a^b} f(x)dx \leq U(f, P) - L(f, P) < \varepsilon$$

and so

$$\left| \overline{\int_a^b} f(x)dx - \underline{\int_a^b} f(x)dx \right| = \overline{\int_a^b} f(x)dx - \underline{\int_a^b} f(x)dx < \varepsilon.$$

Therefore, we have

$$\overline{\int_a^b} f(x)dx = \underline{\int_a^b} f(x)dx$$

and $f \in \mathfrak{R}[a, b]$. This completes the proof $\square$

Example 9.4

Let $f : [a, b] \rightarrow \mathbb{R}$ be the function defined by $f(x) = x$ for every $x \in [a, b]$. By using the definition of the Riemann integral, show that

$$\int_a^b f(x)dx = \frac{1}{2}(b^2 - a^2)$$

Solution. For every $n \geq 1$, let P_n be a partition of $[a, b]$ given by

$$P_n = \left\{ a, a + \frac{b-a}{n}, a + \frac{2(b-a)}{n}, \dots, a + \frac{n(b-a)}{n} = b \right\}.$$

Since $f(x) = x$ is increasing on $[a, b]$, at each subinterval $\left[a + \frac{(i-1)(b-a)}{n}, a + \frac{i(b-a)}{n}\right]$, f has the minimum value $m_i = a + \frac{(i-1)(b-a)}{n}$ and the maximum value $M_i = a + \frac{i(b-a)}{n}$.

Therefore, we have

$$L(f, P_n) = \sum_{i=1}^n m_i \Delta x_i = \sum_{i=1}^n \left(a + \frac{(i-1)(b-a)}{n} \right) \left(\frac{b-a}{n} \right) = a(b-a) + \frac{(b-a)^2}{2} \left(1 - \frac{1}{n} \right).$$

Similarly, we have

$$U(f, P_n) = \sum_{i=1}^n M_i \Delta x_i = \sum_{i=1}^n \left(a + \frac{i(b-a)}{n} \right) \left(\frac{b-a}{n} \right) = a(b-a) + \frac{(b-a)^2}{2} \left(1 + \frac{1}{n} \right)$$

which implies that $f(x) = x$ is Riemann integrable on $[a, b]$ and

$$\int_a^b f(x) dx = \frac{1}{2}(b^2 - a^2)$$

Corollary 9.3

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function. Then, the following are equivalent: (1)

$f \in \mathcal{R}[a, b]$;

(2) There exists a sequence $\{P_n\}$ of partitions of $[a, b]$

$$\lim_{n \rightarrow \infty} (U(f, P_n) - L(f, P_n)) = 0.$$

In this case, we have

$$\lim_{n \rightarrow \infty} L(f, P_n) = \int_a^b f(x) dx = \lim_{n \rightarrow \infty} U(f, P_n).$$

Example 9.5

Let $f : [0, 1] \rightarrow \mathbb{R}$ be the function defined by $f(x) = x$ for every $x \in [0, 1]$. By using the Riemann Integral Test, prove that $f(x) = x$ is Riemann integrable on $[0, 1]$.

Solution. Let $P_n \in \wp[0, 1]$ as in Example 3.4 be a partition of $[a, b]$. Then, we have

$$L(f, P_n) = \frac{n-1}{2n}, \quad U(f, P_n) = \frac{n+1}{2n}.$$

For every $\varepsilon > 0$, if we take a positive integer n_0 such that

$$\frac{1}{\varepsilon} < n_0$$

then we have

$$U(f, P_{n_0}) - L(f, P_{n_0}) = \frac{1}{n_0} < \varepsilon.$$

Therefore, by Theorem (the Riemann Integral Test), f is Riemann integrable on $[0, 1]$.

Theorem 9.6

Every monotone function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable.

• Proof

Assume that f is increasing on $[a, b]$. Since $f(a) \leq f(x) \leq f(b)$ for every $x \in [a, b]$, f is clearly bounded on $[a, b]$. Let $\varepsilon > 0$, and select a positive integer n such that

$$\frac{(f(b) - f(a))(b - a)}{n} < \varepsilon.$$

For a partition

$$P = \{a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b\},$$

where $\Delta x_i = x_i - x_{i-1} = \frac{b-a}{n}$ for every $i \geq 1$, we have

$$U(f, P) = \sum_{i=1}^n M_i \Delta x_i = \sum_{i=1}^n f(x_i) \frac{b-a}{n}$$

and

$$L(f, P) = \sum_{i=1}^n m_i \Delta x_i = \sum_{i=1}^n f(x_{i-1}) \frac{b-a}{n}.$$

Thus, since f is increasing on $[a, b]$, we have

$$U(f, P) - L(f, P) = \frac{b-a}{n} \sum_{i=1}^n |f(x_i) - f(x_{i-1})| = \frac{b-a}{n} (f(b) - f(a)) < \varepsilon.$$

Therefore, by the Riemann Integral Test, f is Riemann integrable on $[a, b]$. Similarly, we can prove this theorem when f is decreasing on $[a, b]$. This completes the proof.

□

Theorem 9.7

Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. Then, $f \in \mathfrak{R}[a, b]$, that is, f is Riemann integrable on $[a, b]$.

• Proof

Since f is a continuous function on $[a, b]$, f is uniformly continuous on $[a, b]$. Thus, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|x - y| < \delta, x, y \in [a, b] \implies |f(x) - f(y)| < \frac{\varepsilon}{b-a}$$

Now, select a sufficiently large positive integer n , and consider a partition $P = \{x_0, x_1, x_2, \dots, x_n\}$ of $[a, b]$ with

$$\Delta x_1 = \Delta x_2 = \dots = \Delta x_{n_0} = \frac{b-a}{n}$$

Further, for each subinterval $I_i = [x_{i-1}, x_i]$ of the partition P_n ($1 \leq i \leq n$), there exist $t_i, u_i \in I_i$ such that

$$M_i = \sup\{f(x) : x \in [x_{i-1}, x_i]\} = f(t_i)$$

and

$$m_i = \inf\{f(x) : x \in [x_{i-1}, x_i]\} = f(u_i)$$

Thus, we have

$$\begin{aligned} 0 \leq U(f, P_n) - L(f, P_n) &= \sum_{i=1}^n (M_i - m_i) \Delta x_i \\ &= \sum_{i=1}^n (f(t_i) - f(u_i)) \Delta x_i \\ &= \sum_{i=1}^{n_0} |f(t_i) - f(u_i)| \left(\frac{b-a}{n} \right) \\ &< \frac{\varepsilon}{b-a} (b-a) = \varepsilon \end{aligned}$$

Therefore, by the Riemann Integral Test, f is Riemann integrable on $[a, b]$. This completes the proof. $\square$

1.2 Properties of the Riemann Integral

In this section, we give the Riemann integrals of the composition and the product of two Riemann integrable functions and some basic properties of the Riemann integral including some algebraic properties of the Riemann integral.

Theorem 9.8

Let $f : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable on $[a, b]$ and $g : [c, d] \rightarrow \mathbb{R}$ be a continuous function on $[c, d]$ with $f([a, b]) \subset [c, d]$. Then, the composition $g \circ f$ is Riemann integrable on $[a, b]$.

• Proof

Let $\varepsilon > 0$. Since g is continuous on $[c, d]$, let

$$M = \sup\{|g(x)| : x \in [c, d]\}$$

and let

$$\varepsilon' = \frac{\varepsilon}{2M + (b - a)}.$$

Also, g is uniformly continuous on $[c, d]$, and there exists $\delta > 0$ such that $0 < \delta < \varepsilon'$ and

$$y, z \in [c, d], |y - z| \leq \delta \implies |g(y) - g(z)| < \varepsilon'.$$

On the other hand, since f is Riemann integrable on $[a, b]$, there exists a partition $P = \{a = x_0, x_1, x_2, \dots, x_n = b\}$ of $[a, b]$ such that

$$U(f, P) - L(f, P) < \delta^2.$$

Now, we show that, for this partition P ,

$$U(g \circ f, P) - L(g \circ f, P) \leq \varepsilon$$

and then $g \circ f$ is Riemann integrable on $[a, b]$. For every $i = 1, 2, \dots, n$, put

$$\begin{aligned} M_i &= \sup\{f(x) : x \in [x_{i-1}, x_i]\}, \\ m_i &= \inf\{f(x) : x \in [x_{i-1}, x_i]\}, \\ M'_i &= \sup\{(g \circ f)(x) : x \in [x_{i-1}, x_i]\}, \\ m'_i &= \inf\{(g \circ f)(x) : x \in [x_{i-1}, x_i]\}. \end{aligned}$$

Now, to be convenient, we separate the indices of the partition P into two disjoint subsets X and Y as follows:

$$X = \{k : M_k - m_k < \delta\}, \quad Y = \{i : M_i - m_i \geq \delta\}.$$

Now, if $i \in X$, that is, $M_i - m_i < \delta$, then, for every $x, x' \in [x_{i-1}, x_i]$, we have

$$|f(x) - f(x')| < \delta$$

and so

$$|(g \circ f)(x) - (g \circ f)(x')| < \varepsilon'$$

which implies that

$$M'_i - m'_i \leq \varepsilon'$$

Thus, we have

$$\sum_{i \in X} (M'_i - m'_i)(x_i - x_{i-1}) \leq \varepsilon'(b - a). \quad (1)$$

But, if $i \in Y$, then we have $\delta \leq M_i - m_i$ and so

$$\begin{aligned} \sum_{i \in Y} (x_i - x_{i-1}) &\leq \frac{1}{\delta} \sum_{i \in Y} (M_i - m_i)(x_i - x_{i-1}) \\ &\leq \frac{1}{\delta} (U(f, P) - L(f, P)) \\ &< \delta < \varepsilon' \end{aligned}$$

Therefore, we have

$$\sum_{i \in Y} (M'_i - m'_i)(x_i - x_{i-1}) \leq 2M\varepsilon'. \quad (2)$$

If we combine (1) and (2) then we have

$$\begin{aligned} U(g \circ f, P) - L(g \circ f, P) &= \sum_{i \in X} (M'_i - m'_i)(x_i - x_{i-1}) + \sum_{i \in Y} (M'_i - m'_i)(x_i - x_{i-1}) \\ &\leq \varepsilon'(b - a) + 2M\varepsilon' = \varepsilon. \end{aligned}$$

Therefore, by the Riemann Integral Test, $g \circ f$ is Riemann integrable on $[a, b]$. This completes the proof. $\square$

Theorem 9.9

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable on $[a, b]$, that is, $f, g \in \mathfrak{R}[a, b]$, and let $c \in \mathbb{R}$. Then, we have the following: (1) $f + g \in \mathfrak{R}[a, b]$ and

$$\int_a^b (f + g)(x) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$$

(2) $cf \in \mathfrak{R}[a, b]$ and

$$\int_a^b cf(x) dx = c \int_a^b f(x) dx$$

(3) $fg \in \mathfrak{R}[a, b]$ (4) If $f(x) \leq g(x)$ for every $x \in [a, b]$, then

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx$$

(5) If $|f| \in \mathfrak{R}[a, b]$ then $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$

• Proof

(1) Since $f, g \in \mathfrak{R}[a, b]$, for every $\varepsilon > 0$, there exist $P_1, P_2 \in \wp[a, b]$ such that

$$U(f, P_1) - L(f, P_1) < \frac{\varepsilon}{2}, \quad U(g, P_2) - L(g, P_2) < \frac{\varepsilon}{2}$$

Put $P = P_1 \cup P_2$. Then, we have $P \in \wp[a, b]$ and

$$\begin{aligned} U(f+g, P) - L(f+g, P) &\leq U(f, P) + U(g, P) - L(f, P) - L(g, P) \\ &= U(f, P) - L(f, P) + U(g, P) - L(g, P) \\ &\leq U(f, P_1) - L(f, P_1) + U(g, P_2) - L(g, P_2) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Therefore, by the Riemann Integral Test, we have $f+g \in \mathfrak{R}[a, b]$. On the other hand, it follows that, for every $P \in \wp[a, b]$,

$$\begin{aligned} L(f, P) + L(g, P) &\leq L(f+g, P) \leq \int_a^b (f+g)(x)dx \\ &\leq U(f+g, P) \leq U(f, P) + U(g, P). \end{aligned} \tag{3}$$

Observe that

$$U(f, P) \leq U(f, P_1) < L(f, P_1) + \frac{\varepsilon}{2} \leq \int_a^b f(x)dx + \frac{\varepsilon}{2}$$

and

$$U(g, P) \leq U(g, P_2) < L(g, P_2) + \frac{\varepsilon}{2} \leq \int_a^b g(x)dx + \frac{\varepsilon}{2}$$

Thus, by (3), we have

$$\int_a^b (f+g)(x)dx \leq U(f+g, P) \leq U(f, P) + U(g, P) < \int_a^b f(x)dx + \int_a^b g(x)dx + \varepsilon$$

Since $\varepsilon > 0$ is arbitrary, we have

$$\int_a^b (f+g)(x)dx \leq \int_a^b f(x)dx + \int_a^b g(x)dx \tag{4}$$

Similarly, we have

$$\int_a^b (f+g)(x)dx \geq L(f+g, P) \geq L(f, P) + L(g, P) \geq U(f, P) - \frac{\varepsilon}{2} + U(g, P) - \frac{\varepsilon}{2} > \int_a^b f(x)dx + \int_a^b g(x)dx - \varepsilon$$

Which leads to

$$\int_a^b (f+g)(x)dx \geq \int_a^b f(x)dx + \int_a^b g(x)dx \tag{5}$$

Therefore, from (4) and (5), it follows that

$$\int_a^b (f+g)(x)dx = \int_a^b f(x)dx + \int_a^b g(x)dx$$

(2) It is easy to prove (2) by the definition of the Riemann integral. (3) Let $f \in \mathfrak{R}[a, b]$ and $h(x) = x^2$ for all $x \in [a, b]$. Then, we have

$$h \circ f = f^2 \in \mathfrak{R}[a, b].$$

Next, let $f, g \in \mathfrak{R}[a, b]$. Then, by (1) and (2), we have

$$-g, f + g, f - g, (f + g)^2, (f - g)^2 \in \mathfrak{R}[a, b].$$

Therefore, since

$$fg = \frac{1}{4} [(f + g)^2 - (f - g)^2]$$

we have $fg \in \mathfrak{R}[a, b]$. (4) Just consider $H(x) = g(x) - f(x) \geq 0$. (5) Since $g(x) = |x|$ is continuous on $[a, b]$, we have $|f| = g \circ f \in \mathfrak{R}[a, b]$. On the other hand, let $c \in \mathbb{R}$ ($c = \pm 1$) be such that

$$c \int_a^b f(x) dx = \left| \int_a^b f(x) dx \right|$$

Then, since $cf(x) \leq |f(x)|$ for every $x \in [a, b]$, we have

$$\int_a^b cf(x) dx \leq \int_a^b |f(x)| dx$$

Therefore, by (2), we have

$$\left| \int_a^b f(x) dx \right| = c \int_a^b f(x) dx = \int_a^b cf(x) dx \leq \int_a^b |f(x)| dx.$$

This completes the proof. $\square$

From the theorem above, we know that

$$\mathfrak{R}[a, b] = \{f : [a, b] \rightarrow \mathbb{R} : \text{a Riemann integrable function on } [a, b]\}$$

is a vector space.

Theorem 9.10

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function on $[a, b]$, and let a number c with $a < c < b$. Then, the following are equivalent:

1. f is Riemann integrable on $[a, b]$;
2. f is Riemann integrable on both $[a, c]$ and $[c, b]$. In this case, we have

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

• Proof

(2) $\implies$ (1) Suppose that f is Riemann integrable on both $[a, c]$ and $[c, b]$. Then, by the Riemann Integral Test, for every $\varepsilon > 0$, there exist $P_1 \in \wp[a, c]$ and $P_2 \in \wp[c, b]$ such that

$$U(f, P_1) - L(f, P_1) < \frac{\varepsilon}{2} \quad \text{and} \quad U(f, P_2) - L(f, P_2) < \frac{\varepsilon}{2}$$

Let $P = P_1 \cup P_2$. Then, we have

$$U(f, P) - L(f, P) = [U(f, P_1) + U(f, P_2)] - [L(f, P_1) + L(f, P_2)] = [U(f, P_1) - L(f, P_1)] + [U(f, P_2) - L(f, P_2)]$$

Since $\varepsilon > 0$ is arbitrary, by the Riemann Integral Test, f is Riemann integrable on $[a, b]$. (1) $\implies$ (2) Suppose that f is Riemann integrable on $[a, b]$. Then, by the Riemann Integral Test, for every $\varepsilon > 0$, there exists $P_1 \in \wp[a, b]$ such that

$$U(f, P_1) - L(f, P_1) < \varepsilon$$

If $P = P_1 \cup \{c\}$, then P is a refinement of P_1 and so

$$U(f, P) - L(f, P) \leq U(f, P_1) - L(f, P_1) < \varepsilon$$

Let $P_2 = P \cap [a, c]$ and $P_3 = P \cap [c, b]$. Then, for $P = P_2 \cup P_3$, we have

$$U(f, P) = U(f, P_2) + U(f, P_3), \quad L(f, P) = L(f, P_2) + L(f, P_3)$$

and so, from (9.6), it follows that

$$[U(f, P_2) - L(f, P_2)] + [U(f, P_3) - L(f, P_3)] < \varepsilon$$

Thus, we have

$$U(f, P_2) - L(f, P_2) < \varepsilon, \quad U(f, P_3) - L(f, P_3) < \varepsilon$$

Since $\varepsilon > 0$ is arbitrary, by the Riemann Integral Test, f is Riemann integrable on both $[a, c]$ and $[c, b]$. Finally, we prove

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

In fact, from (9.7), it follows that

$$\int_a^b f(x)dx \leq U(f, P) = U(f, P_2) + U(f, P_3) < L(f, P_2) + L(f, P_3) + 2\varepsilon \leq \int_a^c f(x)dx + \int_c^b f(x)dx + 2\varepsilon$$

Also, from (9.7), it follows that

$$\int_a^c f(x)dx + \int_c^b f(x)dx \leq U(f, P_2) + U(f, P_3) < L(f, P_2) + L(f, P_3) + 2\varepsilon = L(f, P) + 2\varepsilon \leq \int_a^b f(x)dx + 2\varepsilon$$

Since $\varepsilon > 0$ is arbitrary, it follows from (9.8) and (9.9) that

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx.$$

This completes the proof. □

1.3 The Fundamental Theorems of Calculus

In this section, we prove the Fundamental Theorem of Calculus for a connection between the derivative and the Riemann integral.

Theorem 9.11

Let $f : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable on $[a, b]$, and define a function $F : [a, b] \rightarrow \mathbb{R}$ by

$$F(x) = \int_a^x f(t) dt \quad \text{for every } x \in [a, b]$$

Then, we have the following:

1. F is uniformly continuous on $[a, b]$;
2. If f is continuous at $x_0 \in [a, b]$, then F is differentiable at x_0 and $F'(x_0) = f(x_0)$

• Proof

(1) Since f is bounded on $[a, b]$, there exists a constant $M > 0$ such that

$$|f(t)| \leq M \quad \text{for every } t \in [a, b]$$

Further, for every $x, y \in [a, b]$, we have

$$|F(x) - F(y)| = \left| \int_x^y f(t) dt \right| \leq \int_x^y |f(t)| dt \leq M|y - x|.$$

Therefore, for every $\varepsilon > 0$, if $\delta = \frac{\varepsilon}{M}$, then

$$|y - x| < \delta, x, y \in [a, b] \implies |F(y) - F(x)| < \varepsilon$$

which implies that F is uniformly continuous on $[a, b]$. (2) Let $\varepsilon > 0$. Since f is continuous at $x = x_0$, there exists $\delta > 0$ such that

$$|x - x_0| < \delta, x \in [a, b] \implies |f(x) - f(x_0)| < \varepsilon$$

Thus, if $0 < |x - x_0| < \delta$, then we have

$$\left| \frac{F(x) - F(x_0)}{x - x_0} - f(x_0) \right| = \left| \frac{1}{x - x_0} \left(\int_0^x f(t) dt - \int_0^{x_0} f(t) dt \right) - f(x_0) \right| \leq \frac{1}{|x - x_0|} \int_{x_0}^x |f(t) - f(x_0)| dt < \varepsilon$$

Therefore, F is differentiable at x_0 , and $F'(x_0) = f(x_0)$. This completes the proof. $\square$

Theorem 9.12 (The Fundamental Theorem of Calculus I)

Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function on $[a, b]$, and define a function $F : [a, b] \rightarrow \mathbb{R}$ by

$$F(x) = \int_a^x f(t)dt \quad \text{for every } x \in [a, b].$$

Then, F is differentiable on $[a, b]$, and $F'(x) = f(x)$.

• **Proof**

Since f is continuous on $[a, b]$, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|t - x| < \delta \implies |f(t) - f(x)| < \varepsilon.$$

Let h be a number with $0 < h < \delta$. Then, we have

$$F(x+h) - F(x) = \int_a^{x+h} f(t)dt - \int_a^x f(t)dt = \int_x^{x+h} f(t)dt$$

and so

$$\left| \frac{F(x+h) - F(x)}{h} - f(x) \right| = \left| \frac{1}{h} \int_x^{x+h} f(t)dt - \frac{1}{h} \int_x^{x+h} f(x)dt \right| \leq \frac{1}{h} \int_x^{x+h} |f(t) - f(x)|dt < \varepsilon.$$

Therefore, we have

$$F'_+(x) = \lim_{h \rightarrow 0^+} \frac{F(x+h) - F(x)}{h} = f(x).$$

Similarly, for $-\delta < h < 0$, we have

$$F'_-(x) = \lim_{h \rightarrow 0^-} \frac{F(x+h) - F(x)}{h} = f(x).$$

Since $F'_+(x) = F'_-(x) = f(x)$, F is differentiable on $[a, b]$ and, for all $x \in [a, b]$, $F'(x) = f(x)$. This completes the proof. $\square$

Theorem 9.13 (The Fundamental Theorem of Calculus II)

Let $f : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable on $[a, b]$ and F be differentiable on $[a, b]$. If $F'(x) = f(x)$, then we have

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$$

• **Proof**

Let $P = \{a = x_0, x_1, x_2, \dots, x_n = b\}$ be a partition of $[a, b]$. Then, if we apply Mean Value Theorem to F at each subinterval $I_i = [x_{i-1}, x_i]$ for every $i = 1, 2, \dots, n$,

then it follows that there exists $t_i \in [x_{i-1}, x_i]$ such that

$$F(x_i) - F(x_{i-1}) = f(t_i)(x_i - x_{i-1})$$

Thus, we have

$$F(b) - F(a) = \sum_{i=1}^n (F(x_i) - F(x_{i-1})) = \sum_{i=1}^n f(t_i)(x_i - x_{i-1}).$$

Put

$$M_i = \sup\{f(x) : x \in [x_{i-1}, x_i]\}, \quad m_i = \inf\{f(x) : x \in [x_{i-1}, x_i]\}.$$

Then, we have $m_i \leq f(t_i) \leq M_i$ for every $i = 1, 2, \dots, n$ and so

$$\sum_{i=1}^n m_i(x_i - x_{i-1}) \leq F(b) - F(a) \leq \sum_{i=1}^n M_i(x_i - x_{i-1}).$$

Thus, it follows that, for every $P \in \wp[a, b]$,

$$L(f, P) \leq F(b) - F(a) \leq U(f, P)$$

and so, since

$$\sup\{L(f, P) : P \in \wp[a, b]\} \leq F(b) - F(a) \leq \inf\{U(f, P) : P \in \wp[a, b]\}$$

and $f \in \mathfrak{R}[a, b]$, we have

$$\int_a^b f(x)dx = F(b) - F(a).$$

This completes the proof. □

Example 9.6

Find an example in which if F is not differentiable on $[a, b]$, then Theorem (9.13) is not true.

Solution. Define two functions $f, F : [0, 1] \rightarrow \mathbb{R}$ by $f(x) = 1$ for every $x \in [0, 1]$ and

$$F(x) = \begin{cases} x, & \text{if } x \in [0, 1), \\ 0, & \text{if } x = 1, \end{cases}$$

respectively. Then, f is Riemann integrable on $[0, 1]$ and $\int_0^1 f(x)dx = 1$. On the other hand, we have $F(1) = F(0) = 0$, and, for every $x \in [0, 1)$, $F'(x) = f(x)$, but F is not differentiable at $x = 1$. Thus, the Theorem (9.13) is not true.

Theorem 9.14 (The Generalized Mean Value Theorem for Integral)

Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function on $[a, b]$ and $g : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable on $[a, b]$ with $g(x) \geq 0$ for every $x \in [a, b]$. Then, there exists $c \in [a, b]$ such that

$$\int_a^b f(x)g(x)dx = f(c) \int_a^b g(x)dx.$$

• **Proof**

Since $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$, let

$$M = \sup\{f(x) : x \in [a, b]\}, \quad m = \inf\{f(x) : x \in [a, b]\}$$

Then, it follows that

$$mg(x) \leq f(x)g(x) \leq Mg(x) \quad \text{for every } x \in [a, b]$$

and fg is Riemann integrable on $[a, b]$ and so

$$m \int_a^b g(x)dx \leq \int_a^b f(x)g(x)dx \leq M \int_a^b g(x)dx$$

If $\int_a^b g(x)dx = 0$, then the conclusion follows easily. If $\int_a^b g(x)dx \neq 0$, then it follows from (9.10) that

$$m \leq \frac{\int_a^b f(x)g(x)dx}{\int_a^b g(x)dx} \leq M$$

and so, since f is continuous on $[a, b]$, by the Intermediate Value Theorem, there exists $c \in [a, b]$ such that

$$f(c) = \frac{\int_a^b f(x)g(x)dx}{\int_a^b g(x)dx}$$

that is,

$$\int_a^b f(x)g(x)dx = f(c) \int_a^b g(x)dx$$

This completes the proof. □

Example 9.7

Define two functions $f, g : [-1, 1] \rightarrow \mathbb{R}$ by

$$f(x) = x, \quad g(x) = e^x \quad \text{for every } x \in [-1, 1]$$

Then, apply Theorem (9.14) to the functions f and g on $[-1, 1]$.

Solution. Since f and g are continuous on $[-1, 1]$, they are Riemann integrable on

$[-1, 1]$. Further, $g(x) > 0$ for every $x \in [-1, 1]$ and

$$\int_{-1}^1 f(x)g(x)dx = \frac{2}{e}, \quad \int_{-1}^1 g(x)dx = \frac{e^2 - 1}{e}$$

and so

$$\frac{2}{e^2 - 1} = \frac{\int_{-1}^1 f(x)g(x)dx}{\int_{-1}^1 g(x)dx}$$

Since f is continuous on $[-1, 1]$ and

$$f(-1) = -1 < \frac{2}{e^2 - 1} < 1 = f(1)$$

by the Intermediate Value Theorem, there exists $c \in [-1, 1]$ such that

$$c = f(c) = \frac{2}{e^2 - 1}$$

Therefore, we have

$$f(c) \int_{-1}^1 g(x)dx = \frac{2}{e^2 - 1} \cdot \frac{e^2 - 1}{e} = \frac{2}{e} = \int_{-1}^1 f(x)g(x)dx.$$

Corollary 9.4 (The Mean Value Theorem for Integral)

Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function on $[a, b]$. Then, there exists $c \in [a, b]$ such that

$$\int_a^b f(x)dx = f(c)(b - a)$$

• Proof

In Theorem (9.14), if $g(x) = 1$ for every $x \in [a, b]$, then we have

$$\int_a^b f(x)dx = f(c)(b - a)$$

□

Example 9.8

By using the Mean Value Theorem for Integral, prove the following inequalities:

$$\frac{x}{1+x} < \ln(1+x) < x \quad \text{for every } x > 0$$

Solution. Let $f(x) = \frac{1}{1+x}$ for every $x > 0$. Then, we have

$$\int_0^x \frac{1}{1+t} dt = \ln(1+x)$$

Thus, by Corollary (9.4), there exists $c \in [0, x]$ such that

$$\int_0^x \frac{1}{1+t} dt = f(c)(x-0)$$

Since f is decreasing on $[0, x]$ and

$$\inf\{f(t) : t \in [0, x]\} \leq f(c) \leq \sup\{f(t) : t \in [0, x]\}$$

we have

$$\frac{1}{1+x} < \frac{\ln(1+x)}{x} < 1 \quad \text{for every } x > 0$$

which implies that

$$\frac{x}{1+x} < \ln(1+x) < x \quad \text{for every } x > 0$$

1.3.1 The Substitution Theorem and Integration by Parts

In this section, we prove the Substitution Theorem and the Integration by Parts as techniques of integration, which are based on the Fundamental Theorems of Calculus.

Theorem 9.15 (The Substitution Theorem)

Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function on $[a, b]$ and $g : [c, d] \rightarrow [a, b]$ be differentiable on $[c, d]$. If $g(c) = a$ and $g(d) = b$, then we have

$$\int_a^b f(x) dx = \int_c^d f(g(t))g'(t) dt$$

• Proof

For every $x \in [a, b]$, define a function $F : [a, b] \rightarrow \mathbb{R}$ by

$$F(x) = \int_a^x f(t) dt$$

Then, by Theorem (The Fundamental Theorem of Calculus I), F is differentiable on $[a, b]$. Thus, we have

$$(F(g(t)))' = F'(g(t))g'(t) = f(g(t))g'(t).$$

Therefore, by Theorem (The Fundamental Theorem of Calculus II), we have

$$\int_c^d f(g(t))g'(t) dt = F(g(d)) - F(g(c)) = F(b) - F(a) = \int_a^b f(x) dx$$

This completes the proof. □

Example 9.9

Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function on $[0, 1]$. Evaluate the following:

$$\int_{-1}^1 x f(x^2) dx$$

Solution. Let $\phi(x) = x^2$ for every $x \in [-1, 1]$. Then, since we have $\phi([-1, 1]) = [0, 1]$ and f is continuous on $[0, 1]$, we have

$$\int_{-1}^1 f(x^2) dx = \frac{1}{2} \int_{-1}^1 f(\phi(x)) \phi'(x) dx = \frac{1}{2} \int_1^1 f(t) dt = 0$$

Theorem 9.16 (The Integration by Parts)

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be differentiable on $[a, b]$ and f', g' be Riemann integrable on $[a, b]$. Then, we have

$$\int_a^b f(x) g'(x) dx = f(b)g(b) - f(a)g(a) - \int_a^b f'(x)g(x) dx$$

• **Proof**

Since f and g are differentiable on $[a, b]$, fg is also differentiable on $[a, b]$ and so, for every $x \in [a, b]$,

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x).$$

Since f and g are continuous on $[a, b]$, they are Riemann integrable on $[a, b]$. Since f' and g' are Riemann integrable on $[a, b]$, $(fg)' = f'g + fg'$ is also Riemann integrable on $[a, b]$. Therefore, by Theorem (The Fundamental Theorem of Calculus I), we have

$$\int_a^b (f(x)g(x))' dx = f(b)g(b) - f(a)g(a)$$

and so

$$\int_a^b f'(x)g(x) dx + \int_a^b f(x)g'(x) dx = f(b)g(b) - f(a)g(a),$$

which implies that

$$\int_a^b f(x)g'(x) dx = f(b)g(b) - f(a)g(a) - \int_a^b f'(x)g(x) dx.$$

This completes the proof. □

Example 9.10

Evaluate the following:

$$\int_0^{\pi/2} \cos^3 x dx$$

Solution. Now, we have

$$\begin{aligned} \int_0^{\pi/2} \cos^3 x dx &= \int_0^{\pi/2} \cos^2 x (\cos x dx) \\ &= \int_0^{\pi/2} (1 - \sin^2 x)(\cos x dx). \end{aligned}$$

Let $u = \sin x$. Then, we have $du = \cos x dx$. If $x = 0$, then $u = 0$. If $x = \frac{\pi}{2}$, then $u = 1$. Therefore, we have

$$\begin{aligned} \int_0^{\pi/2} \cos^3 x dx &= \int_0^1 (1 - u^2) du \\ &= \left[u - \frac{1}{3} u^3 \right]_0^1 \\ &= 1 - \frac{1}{3} = \frac{2}{3}. \end{aligned}$$

Theorem 9.17 (Luxemburg Monotone Convergence Theorem)

Let $\{f_n(x)\}$ be a decreasing sequence of bounded functions on $[a, b]$, with $a < b$, which converges pointwise to 0 on $[a, b]$. Then

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = 0.$$

• **Proof**

The lower integral of any function is well defined provided the function is bounded. Therefore for any bounded function $f(x)$, and by definition of the lower integral, for any $\varepsilon > 0$, there exists a step function $L(x) \leq f(x)$ such that

$$\int_a^b (f(x) - L(x)) dx < \frac{\varepsilon}{2}$$

Also note the existence of a continuous function $c(x) \leq L(x)$ such that

$$\int_a^b (L(x) - c(x)) dx < \frac{\varepsilon}{2}$$

Putting all this together, we conclude that for any bounded function $f(x)$, and for any $\varepsilon > 0$, there exists a continuous function $c(x) \leq f(x)$ such that

$$\int_a^b (f(x) - c(x)) dx < \varepsilon$$

Note that if $f(x) \geq 0$, then the construction of $c(x)$ will be done to have $c(x) \geq 0$ as well. Back to our claim. Let $\varepsilon > 0$. Then there exists a positive continuous function $c_1(x) \leq f_1(x)$ such that

$$\int_a^b (f_1(x) - c_1(x))dx < \frac{\varepsilon}{2^2}$$

Since $\min(c_1(x), f_2(x))$ is bounded and positive, there exists a positive continuous function $c_2(x) \leq \min(c_1(x), f_2(x))$ such that

$$\int_a^b (\min(c_1(x), f_2(x)) - c_2(x))dx < \frac{\varepsilon}{2^3}$$

Since $f_2(x) - c_2(x) \leq f_2(x) - \min(c_1(x), f_2(x)) + \min(c_1(x), f_2(x)) - c_2(x)$, and $f_2(x) \leq f_1(x)$, we get

$$\int_a^b (f_2(x) - c_2(x))dx \leq \int_a^b (f_1(x) - c_1(x))dx + \int_a^b (\min(c_1(x), f_2(x)) - c_2(x))dx < \frac{\varepsilon}{2^2} + \frac{\varepsilon}{2^3}.$$

By the induction argument, a similar construction will lead to the existence of a decreasing sequence of positive continuous functions $\{c_n(x)\}$ such that $c_n(x) \leq f_n(x)$ and

$$\int_a^b (f_n(x) - c_n(x))dx < \frac{\varepsilon}{2^2} + \frac{\varepsilon}{2^3} + \cdots + \frac{\varepsilon}{2^{n+1}} < \frac{\varepsilon}{2}.$$

Since $\{f_n(x)\}$ converges pointwise to 0 on $[a, b]$ this will force the sequence $\{c_n(x)\}$ to also converge pointwise to 0 on $[a, b]$. Dini's theorem will imply that $\{c_n(x)\}$ converges uniformly to 0 on $[a, b]$. So there exists $n_0 \geq 1$ such that for any $n \geq n_0$ we have $c_n(x) \leq \frac{\varepsilon}{2(b-a)}$ for any $x \in [a, b]$. Hence

$$\int_a^b f_n(x)dx \leq \int_a^b (f_n(x) - c_n(x))dx + \int_a^b c_n(x)dx < \frac{\varepsilon}{2} + \frac{\varepsilon}{2(b-a)}(b-a) = \varepsilon$$

whenever $n \geq n_0$. This finishes the proof of our claim. $\square$

Theorem 9.18 (Monotone Convergence Theorem)

Let $\{f_n(x)\}$ be a decreasing sequence of Riemann integrable functions on $[a, b]$ which converges pointwise to a Riemann integrable function $f(x)$. Then

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x)dx = \int_a^b f(x)dx$$

• Proof

Since $f(x)$ is Riemann integrable, $\{f_n(x) - f(x)\}$ is a sequence of Riemann integrable functions which decreases to 0. Obviously they are all bounded functions.

The Luxemburg Monotone Convergence Theorem will imply

$$\lim_{n \rightarrow \infty} \int_a^b (f_n(x) - f(x)) dx = 0.$$

But $\int_a^b (f_n(x) - f(x)) dx = \int_a^b (f_n(x) - f(x)) dx$, hence

$$\lim_{n \rightarrow \infty} \int_a^b (f_n(x) - f(x)) dx = 0$$

which implies

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b f(x) dx$$

□

Theorem 9.19 (Arzelà Theorem)

Let $\{f_n(x)\}$ be a sequence of Riemann integrable functions on $[a, b]$ which converges pointwise to a Riemann integrable function $f(x)$. Assume that there exists $M > 0$ such that $|f_n(x)| \leq M$ for any $x \in [a, b]$ and $n \geq 1$. Then

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b f(x) dx$$

• Proof

Since $f(x)$ is Riemann integrable, $\{f_n(x) - f(x)\}$ is a sequence of Riemann integrable functions which decreases to 0. Obviously they are all bounded functions. The Luxemburg Monotone Convergence Theorem will imply

$$\lim_{n \rightarrow \infty} \int_a^b (f_n(x) - f(x)) dx = 0.$$

But $\int_a^b (f_n(x) - f(x)) dx = \int_a^b (f_n(x) - f(x)) dx$, hence

$$\lim_{n \rightarrow \infty} \int_a^b (f_n(x) - f(x)) dx = 0$$

which implies

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b f(x) dx$$

□

Lemma 9.20 (Fatou)

Let $\{f_n(x)\}$ be a sequence of Riemann integrable functions on $[a, b]$ which converges pointwise to a Riemann integrable function $f(x)$. Then

$$\int_a^b f(x)dx \leq \liminf_{n \rightarrow \infty} \int_a^b f_n(x)dx$$

• **Proof**

If $\liminf_{n \rightarrow \infty} \int_a^b f_n(x)dx = \infty$, then the conclusion is obvious. Assume that $\liminf_{n \rightarrow \infty} \int_a^b f_n(x)dx < \infty$. Then there exists a subsequence $\{f_{n_k}\}$ of $\{f_n\}$ such that

$$\lim_{n_k \rightarrow \infty} \int_a^b f_{n_k}(x)dx = \liminf_{n \rightarrow \infty} \int_a^b f_n(x)dx$$

Clearly the subsequence $\{f_{n_k}\}$ also converges pointwise to $f(x)$. Set $h_{n_k}(x) = \inf_{n \geq n_k} f_n(x)$. Then $\{h_{n_k}(x)\}$ also converges pointwise to $f(x)$ and is increasing. It is easy to see that this sequence is bounded. The Luxemburg Monotone Convergence Theorem applied to $\{f(x) - h_{n_k}(x)\}$ will easily imply that

$$\lim_{n \rightarrow \infty} \int_a^b h_{n_k}(x)dx = \int_a^b f(x)dx$$

Since $h_{n_k}(x) \leq f_{n_k}(x)$, we get

$$\int_a^b f(x)dx \leq \lim_{n \rightarrow \infty} \int_a^b f_{n_k}(x)dx = \liminf_{n \rightarrow \infty} \int_a^b f_n(x)dx$$

□